

LESSON 7-7 • SESSION 2 • PRACTICE • FORM A

Designing Polygons with Scale

Learning target – I can use a scale to find a missing dimension of a figure on the coordinate plane. • 6.G.A.3

REMEMBER

- A **scale** tells how long one grid unit is in real life: 1 unit = 4 m.
- Change each side to real length **first** (units $\times$ scale). Then find perimeter or area.
- Missing side = **area $\div$ known side**. Area uses square units (ft^2 , m^2).

WORKED EXAMPLE

A plan uses 1 unit = 10 m. A field must cover 2,400 m^2 and is 4 units wide. Find its real length.

- Step 1** Real width: $4 \times 10 = 40$ m.
Step 2 Real length: $2,400 \div 40 = 60$ m.
Step 3 On the plan: $60 \div 10 = 6$ units long.
Answer 60 m long (6 units on the plan)

WORDS **scale** (*escala*) • **area** (*área*) • **dimensions** (*dimensiones*)

PRACTICE

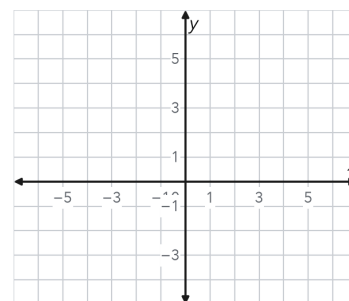
show your work • label units

1. On a plan, 1 grid unit = 6 ft. Change each measure.

- a. 4 units = _____ ft b. 9 units = _____ ft
 c. 42 ft = _____ units d. 90 ft = _____ units

2. **Courtyard.** Three corners of a rectangular courtyard are $(-5, 6)$, $(6, 6)$ and $(6, -3)$. Each unit is 10 feet.

- a. Plot the corners. Name the fourth corner. _____
 b. Width and height in feet _____
 c. Area _____ ft^2



3. **Parking Pad.** Parking pad corners: $(-7, 5)$, $(5, 5)$, $(5, -4)$ and $(-7, -4)$. Each unit is 3 feet. Find the area in square feet.

Hint Change both sides to feet before you multiply.

Show your work.

Answer _____ ft^2

4. **Using the Scale.** On a plan, 1 unit = 25 feet. A rectangle is 6 units by 4 units. Find the real perimeter.

Show your work.

Answer _____ ft

5. **Missing Dimension.** A rectangular lot must cover 30,000 ft^2 . One side is 150 ft. Find the other side.

Hint Divide the area by the known side.

Show your work.

Answer _____ ft

- 6. Fence Cost.** Yard corners: $(-4, 3)$, $(8, 3)$, $(8, -5)$ and $(-4, -5)$. Each unit is 4 feet. Fencing costs \$6 per foot. Find the fence cost.

Hint Change both sides to feet, add all four, then multiply by 6.

Show your work.

Cost _____

- 7. Pen Design.** Mara wants a pen with an area of 384 ft^2 . One side runs from $(-4, 6)$ to $(8, 6)$. Each unit is 2 feet. Find the other side.

Hint Change the known side to feet first.

Show your work.

Answer _____ ft

- 8. Playground Design.** A plan uses 1 unit = 15 feet. A playground must cover $5,400 \text{ ft}^2$. On the plan it is 8 units wide.

- a. Real width _____ ft
- b. Real length _____ ft
- c. Length on the plan _____ units
- d. Real perimeter _____ ft

FIND THE ERROR

explain the mistake, then fix it

- 9. Talia's Work.** A side runs from $(-3, 4)$ to $(9, 4)$. Each unit is 6 feet. Talia wrote the side length as 12 feet. Explain her mistake. Then find the correct length.

TEST PRACTICE

choose the best answer

- 10.** An area of $1,200 \text{ ft}^2$ with one side of 40 ft has another side of –
A 30 ft **B** 1,160 ft **C** 48,000 ft **D** 20 ft
- 11.** If 1 unit represents 20 feet, then 5 units represent –
A 4 feet **B** 25 feet **C** 100 feet **D** 520 feet

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

16 G	20 J	45 N	500 E	972 D	\$960 I	\$160 Z	32 X	108 Q	200 S
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#3	#4	#5	#6	#7	#8b
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____